Talagrand's discrete-convexity conjecture

Rewritten for readability from the OpenAI manuscript of the same title

September 23, 2026

Abstract

We prove Talagrand's discrete-convexity conjecture. There is a universal integer k such that, whenever an arbitrary family has Bernoulli product measure at least $1 - 1/k$, the sets not contained in a union of k members admit a cover of total cost at most $1/2$ at the same density.

1 Introduction

1.1 Definitions and the main result

Let $[N] = \{1, \dots, N\}$. For $p \in (0,1)$, let μ_p be the probability measure on $2^{[N]}$ under which each element of $[N]$ is present independently with probability p. For a family $\mathcal{D} \subseteq 2^{[N]}$ and an integer $k \geq 1$, the *exceptional family* is

$$E_k(\mathcal{D}) = \{S \subseteq [N] : S \nsubseteq D_1 \cup \cdots \cup D_k \text{ for all } D_1, \dots, D_k \in \mathcal{D}\}. \tag{1.1}$$

The D_j may repeat and need not be disjoint.

A family $\mathcal{G} \subseteq 2^{[N]}$ *covers* a family $\mathcal{A}$ if every $S \in \mathcal{A}$ contains some $I \in \mathcal{G}$; members of $\mathcal{G}$ are called *generators*. For $\rho \in (0,1)$, the *cost* of $\mathcal{G}$ at density ρ is

$$c_\rho(\mathcal{G}) = \sum_{I \in \mathcal{G}} \rho^{|I|}.$$

We say $\mathcal{A}$ is *p-small* if it has a cover $\mathcal{G}$ with $c_p(\mathcal{G}) \leq 1/2$. Since $\mu_p\{S : I \subseteq S\} = p^{|I|}$, a union bound over $\mathcal{G}$ then gives $\mu_p(\mathcal{A}) \leq 1/2$; smallness is an explicit certificate of this. The empty family is covered by $\mathcal{G} = \varnothing$ at cost 0.

Theorem 1.1. *Set $k = 2^{75}$. For every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) \geq 1 - \frac{1}{k} \quad \Longrightarrow \quad E_k(\mathcal{D}) \text{ is } p\text{-small.}$$

The constant is independent of N, p, and $\mathcal{D}$. The family $\mathcal{D}$ is arbitrary; no monotonicity is assumed.

1.2 Proof at a glance

The argument is a chain: Lemma 2.1 ⇒ Proposition 3.1 ⇒ Theorem 1.1.

(A) **A cover at a constant fraction of the density (Proposition 3.1).** If $\mu_q(\mathcal{F}) \geq 1/2$, then $E_{32}(\mathcal{F})$ is $(q/2^{70})$-small. To $\mathcal{F}$ we attach weights $w(U) \geq 0$, $U \subseteq [N]$, with two properties:

(i) *global bound:* $\sum_U q^{|U|} w(U) \le 1$ (Parseval in the biased Fourier basis);

(ii) *local bound:* every $S \in E_{32}(\mathcal{F})$ has $\sum_{\varnothing \ne U \subseteq S} w(U)^{16} \ge 2^{-32}$. This comes from a signed product measure on $32 \times N$ arrays that vanishes unless the union of the rows contains S.

Sets U of large weight are cheap generators by (i). The other sets are sorted into bins by size r and dyadic weight 2^{-h}. Fewer than 2^{12h} sets of weight at most 2^{-h} contribute less than 2^{-4h} to the sum in (ii). So an exceptional set containing no heavy U must contain at least 2^{12h} members of some bin.

(B) **The covering lemma (Lemma 2.1).** This is purely combinatorial and handles one bin: if an r-uniform family $\mathcal{H}$ has $|\mathcal{H}|\rho^r \le 2^{h+1}$, the sets containing at least 2^{12h} members of $\mathcal{H}$ can be covered at cost $3 \cdot 2^{-h}$. By (i), each bin satisfies the hypothesis. The cover is built by a random construction.

(C) **Back to density p (Section 4).** Let $L = 2^{70}$. We couple L random sets, each with law μ_p, so that their union has law μ_{Lp}. Applying (A) with $q = Lp$ to the family $\mathcal{F}$ of L-fold unions of members of $\mathcal{D}$ gives cost density $q/L = p$, and $E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D})$. When $Lp \ge 1$, a second coupling shows $E_k(\mathcal{D}) = \varnothing$.

Lemma 2.1 is independent of the rest, and (i) and (ii) are independent of each other. The by-products in Section 1.3 are not used in the proof of Theorem 1.1.

1.3 Context, notation, and two by-products

Context (not needed for the proofs). The name reflects the analogy with Talagrand's Gaussian convexity question [11, Conjecture 2.1]: can a bounded number of sums of a compact balanced set of sufficiently large Gaussian measure contain a convex set of Gaussian measure at least $1/2$, uniformly in the dimension? In the discrete problem unions replace sums. Theorem 1.1 is the conjecture formulated in [11, Conjecture 7.1]; it appears as Research Problem 13.3.2 in [12], whose Proposition 13.3.3 explains the connection to positive selector processes. Hua, Song, and Tudose resolve the Gaussian problem and derive a discrete conclusion with three unions and covering density p^C [5, Corollaries 1.3–1.4]. Ascoli, He, Park, and Talagrand give a threshold reformulation and prove cases for graph-containment properties [1, Proposition 1.5 and Theorems 1.10–1.13]. Talagrand also introduced a fractional version of smallness [11, Sections 6 and 7] (defined in the proof of Corollary 1.2). Li resolves the fractional problem [6, Theorem 1.9]. Park obtains an integral cover of $E_2(\mathcal{D})$ at density $p/(C \max\{1, \log\log(1/p)\})$ whenever $\mu_p(\mathcal{D}) > 1/2$ [9, Corollary 3.5]. On methods: with two rows, the signed measure of Section 3 is Li's signed kernel multiplied by the Bernoulli masses [6, Section 2]; Park notes the extension to an even number of unions [9, Corollary 3.5, footnote 3]; antecedents are the signed product operators and biased Fourier analysis of [4, Section 2.2] and [2, Sections 2.2 and 4.1]. Random unions also enter the covering constructions of [3, Sections 1.1 and 2]; Lemma 2.1 is proved in full here.

Notation changes. Three symbols of the source are renamed to avoid clashes: the selector class T becomes $\mathcal{T}$ (since $T = 2^{4h}$ in Section 2), the signed column mass R becomes $\widetilde{P}$ (since R_j are random sets in Section 4), and the parameter t of Case 2 becomes τ (since t is a tuple length in Section 2). The source's figure illustrating the coupling is omitted.

By-product 1: positive selectors. For a nonempty $\mathcal{T} \subseteq [0,\infty)^N$ define

$$\phi(S) = \sup_{t\in\mathcal{T}} \sum_{i\in S} t_i, \qquad M = \mathbb{E}_{\mu_p}\phi,$$

and suppose $0 < M < \infty$. Then $\{S : \phi(S) \geq k^2 M\}$ is p-small for $k = 2^{75}$. This recovers, through the connection of [11, Section 7], the positive-selector conclusion of Park and Pham [10, Theorem 1.2].

Proof. Let $\mathcal{D} = \{D : \phi(D) < kM\}$. Markov's inequality gives $\mu_p\{\phi \geq kM\} \leq 1/k$, so $\mu_p(\mathcal{D}) \geq 1 - 1/k$. Because all $t_i \geq 0$, ϕ is monotone and $\phi(A \cup B) \leq \phi(A) + \phi(B)$. Hence if $S \subseteq D_1 \cup \cdots \cup D_k$ with all $D_j \in \mathcal{D}$, then $\phi(S) \leq \sum_j \phi(D_j) < k^2 M$. So $\{\phi \geq k^2 M\} \subseteq E_k(\mathcal{D})$, and the cover of $E_k(\mathcal{D})$ from Theorem 1.1 covers it. □

By-product 2: two unions. Combining Li's result with the companion constant-factor rounding theorem gives the following.

Corollary 1.2 (Two unions at constant density loss)**.** *There is a universal constant $C \geq 1$ such that, for every integer $N \geq 1$, every $p \in (0,1)$, and every family $\mathcal{D} \subseteq 2^{[N]}$,*

$$\mu_p(\mathcal{D}) > \frac{1}{2} \quad \Longrightarrow \quad E_2(\mathcal{D}) \text{ is } (p/C)\text{-small.}$$

Proof. The proof rests on three external statements, quoted as the source states them; I could not recheck them against the cited papers.

Step 1. If $E_2(\mathcal{D}) = \varnothing$, the empty cover suffices. Otherwise $E_2(\mathcal{D})$ is a nonempty proper increasing family. It is increasing because if $S \notin D_1 \cup D_2$ then no superset of S is contained in $D_1 \cup D_2$. It is proper because $\mathcal{D} \neq \varnothing$ (as $\mu_p(\mathcal{D}) > 1/2$), so $\varnothing \notin E_2(\mathcal{D})$.

Step 2. A *fractional cover* of $\mathcal{A}$ of cost at most $1/2$ at density ρ is a choice of coefficients $a_I \geq 0$ with

$$\sum_{I\subseteq S} a_I \geq 1 \quad (S \in \mathcal{A}), \qquad \sum_{I\subseteq[N]} \rho^{|I|} a_I \leq \frac{1}{2}.$$

By Li's result [6, Corollary 3.1] in Park's fractional-cover formulation [9, Theorem 3.4], $E_2(\mathcal{D})$ has such a fractional cover at density $p/2$.

Step 3. Replace each a_I by $\min\{a_I, 1\}$. The cost does not increase. Coverage is preserved: if some $I \subseteq S$ had $a_I > 1$, it now contributes 1; otherwise nothing changed.

Step 4. The companion expectation-threshold theorem [8, Theorem 1.1] then gives an integral cover of cost at most $1/2$ at density $(p/2)/(25 \cdot 512^4)$. Thus $C = 50 \cdot 512^4$ suffices. □

Theorem 1.1 instead keeps the original cost parameter p and allows a larger universal number of unions.

2 An elementary covering lemma

Lemma 2.1. *Let X be a finite set, let $0 < \rho < 1$, and let r, h be integers with $1 \leq r \leq h$ and $h \geq 64$. Suppose that $\mathcal{H} \subseteq \binom{X}{r}$ satisfies*

$$|\mathcal{H}|\rho^r \leq 2^{h+1}.$$

Then the family

$$\left\{S \subseteq X : |\{e \in \mathcal{H} : e \subseteq S\}| \geq 2^{12h}\right\}$$

has a cover $\mathcal{G} \subseteq 2^X$ with

$$c_\rho(\mathcal{G}) \leq 3 \cdot 2^{-h}.$$

Members of $\mathcal{H}$ are called *edges*, and the family to be covered is the *target family*. The lemma is used once, in Step 5 of the proof of Proposition 3.1.

Plan. We build $\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2$. The deterministic part $\mathcal{C}_0$ consists of subsets lying in many edges. $\mathcal{C}_1$ consists of unions of randomly sampled pairs of the remaining edges. $\mathcal{C}_2$ consists of unions of long tuples of edges none of whose pairs was sampled. Every sampling outcome gives a cover; we bound the expected cost and pick a good outcome.

Proof. Put

$$m = 2^{12h}, \qquad T = 2^{4h}, \qquad \pi = 2^{-8h}.$$

For $I \subseteq X$ with $|I| \le r$, define its *weighted degree*

$$d(I) = |\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|}.$$

Step 1: generators of large weighted degree. Let $\mathcal{C}_0 = \{I \subseteq X : |I| \le r,\ d(I) > T\}$. Each edge has 2^r subsets, so exchanging the order of summation gives

$$\sum_{\substack{I \subseteq X \\ |I| \le r}} d(I)\rho^{|I|} = \sum_{e \in \mathcal{H}} \sum_{I \subseteq e} \rho^r = 2^r |\mathcal{H}| \rho^r.$$

Every $I \in \mathcal{C}_0$ has $\rho^{|I|} < T^{-1} d(I) \rho^{|I|}$. Summing, and using the hypothesis on $|\mathcal{H}|$ and $r \le h$,

$$c_\rho(\mathcal{C}_0) \le T^{-1} 2^r |\mathcal{H}| \rho^r \le 2^{-4h+r+h+1} = 2^{r+1-3h} \le 2^{1-2h}. \tag{2.1}$$

Step 2: regular edges and the extension bound. Call an edge *regular* if it contains no member of $\mathcal{C}_0$, and let $\mathcal{H}'$ be the family of regular edges. If $I \subseteq e \in \mathcal{H}'$, then $|I| \le r$ and $I \notin \mathcal{C}_0$, so

$$d(I) \le T. \tag{2.2}$$

Claim. For every $B \subseteq X$,

$$\sum_{e \in \mathcal{H}'} \rho^{|e \setminus B|} \le T \sum_{i=0}^{r} \binom{|B|}{i}. \tag{2.3}$$

To see this, sort the regular edges by $I = e \cap B$. This is a subset of B of size at most r, so there are at most $\sum_{i=0}^{r} \binom{|B|}{i}$ classes. Fix a nonempty class with intersection I. Each of its edges contains I and has $|e \setminus B| = r - |I|$, so the class contributes at most

$$|\{e \in \mathcal{H} : I \subseteq e\}|\, \rho^{r-|I|} = d(I) \le T,$$

by (2.2).

Step 3: sampled pairs. Independently for each ordered pair $(e, e') \in (\mathcal{H}')^2$, diagonal pairs included, sample the pair with probability π. Let $\mathcal{C}_1$ be the set of unions $e \cup e'$ of sampled pairs. Each member of $\mathcal{C}_1$ arises from at least one sampled pair, so $c_\rho(\mathcal{C}_1) \le \sum_{(e,e') \text{ sampled}} \rho^{|e \cup e'|}$. Taking expectations and writing $\rho^{|e \cup e'|} = \rho^r \rho^{|e' \setminus e|}$,

$$\begin{aligned} \mathbb{E}[c_\rho(\mathcal{C}_1)] &\le \pi \sum_{e \in \mathcal{H}'} \rho^r \sum_{e' \in \mathcal{H}'} \rho^{|e' \setminus e|} \le \pi |\mathcal{H}'| \rho^r\, 2^r T \\ &\le 2^{-8h+h+1+r+4h} = 2^{r+1-3h} \le 2^{1-2h}. \end{aligned} \tag{2.4}$$

The second inequality is (2.3) with $B = e$, where $\sum_{i=0}^{r} \binom{r}{i} = 2^r$.

Step 4: residual generators; $\mathcal{G}$ is a cover for every outcome. For every ordered tuple $(e_1, \dots, e_m)$ of distinct regular edges, put $e_1 \cup \cdots \cup e_m$ into $\mathcal{C}_2$ if none of the m^2 pairs (e_i, e_j), $1 \le i, j \le m$, was sampled. Let $\mathcal{G} = \mathcal{C}_0 \cup \mathcal{C}_1 \cup \mathcal{C}_2$.

Let S be in the target family and suppose S contains no member of $\mathcal{C}_0$. Then every edge contained in S is regular, since a non-regular edge contains a member of $\mathcal{C}_0$. So S contains m distinct regular edges $e_1, \dots, e_m$. If some pair (e_i, e_j) was sampled, then $e_i \cup e_j \in \mathcal{C}_1$ lies in S. Otherwise $e_1 \cup \cdots \cup e_m \in \mathcal{C}_2$ lies in S.

Step 5: expected cost of $\mathcal{C}_2$. For $0 \le t \le m$ let

$$W_t = \sum_{\substack{(e_1,\dots,e_t)\in(\mathcal{H}')^t \\ e_1,\dots,e_t \text{ distinct}}} \rho^{|e_1\cup\cdots\cup e_t|}, \qquad W_0 = 1.$$

(a) Recursion. Fix a tuple of length $t < m$ with union B, so $|B| \le rt \le rm$. For a regular edge e, $\rho^{|B\cup e|} = \rho^{|B|}\rho^{|e\setminus B|}$. Summing over *all* regular e (which only enlarges the sum over admissible extensions) and using (2.3), the extensions of this tuple have total weight at most

$$\rho^{|B|}\, T \sum_{i=0}^{r} \binom{|B|}{i} \le \rho^{|B|}\, T(1+rm)^r.$$

Here $\sum_{i=0}^{r} \binom{n}{i} \le (1+n)^r$: order the n-set, list each subset of size at most r in increasing order, and pad to length r with a new symbol; this is an injection into $(n+1)^r$ lists. Summing over tuples of length t gives $W_{t+1} \le T(1+rm)^r W_t$, hence

$$W_m \le [T(1+rm)^r]^m.$$

(b) Survival probability. For a tuple of m distinct regular edges, the m^2 ordered pairs (e_i, e_j) are distinct elements of $(\mathcal{H}')^2$, so they are sampled independently. The tuple contributes to $\mathcal{C}_2$ with probability

$$(1-\pi)^{m^2} \le \exp(-\pi m^2) \le 2^{-\pi m^2}.$$

(c) Expectation. Each member of $\mathcal{C}_2$ arises from at least one contributing tuple, so by linearity

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le W_m\, 2^{-\pi m^2} \le [T(1+rm)^r]^m\, 2^{-\pi m^2}.$$

(d) Numerics. We have $\pi m = 2^{4h}$. Since $r \le h$ and $h+1 \le 2^h$, $1 + rm \le (h+1)2^{12h} \le 2^{13h}$, so $T(1+rm)^r \le 2^{4h+13rh}$. Moreover

$$4h + 13rh + 1 \le 4h + 13h^2 + 1 \le 18h^2 \le 18 \cdot 2^{2h} \le 2^{4h} = \pi m,$$

using in turn $r \le h$, $4h + 1 \le 5h^2$, $h \le 2^h$, and $18 \le 2^{2h}$. Therefore

$$\mathbb{E}[c_\rho(\mathcal{C}_2)] \le 2^{(4h+13rh)m - \pi m^2} \le 2^{(\pi m - 1)m - \pi m^2} = 2^{-m}. \tag{2.5}$$

Step 6: conclusion. By (2.1), (2.4), and (2.5),

$$\mathbb{E}[c_\rho(\mathcal{G})] \le 2^{2-2h} + 2^{-m} \le 3 \cdot 2^{-h},$$

since $2^{2-2h} \le 2^{-h}$ and $m \ge h$. The sample space is finite, so some outcome has cost at most the expectation, and by Step 4 it is a cover. □

3 Weights for exceptional sets

Proposition 3.1. *Let $L = 2^{70}$. For every integer $N \geq 1$, every $q \in (0,1)$, and every $\mathcal{F} \subseteq 2^{[N]}$ satisfying $\mu_q(\mathcal{F}) \geq 1/2$, the family $E_{32}(\mathcal{F})$ is (q/L)-small.*

The proposition is used once, in Case 1 of Section 4.

Plan. Steps 1–3 establish the bounds (i) and (ii) of the overview. Steps 4–6 build a cover of cost below $1/2$ from heavy sets and from Lemma 2.1 applied to bins. Step 7 shows that a set avoiding all generators violates (ii).

Proof. Identify subsets of $[N]$ with indicator vectors in $\{0,1\}^N$ and write $f = \mathbf{1}_{\mathcal{F}}$. For $U \subseteq [N]$, $z \in \{0,1\}^U$ and $y \in \{0,1\}^{[N]\setminus U}$, let $(z,y) \in \{0,1\}^N$ be the vector equal to z on U and to y elsewhere.

Step 0: the weights. For $U \subseteq [N]$ define

$$b(U) = (1-q)^{|U|} \sum_{z \in \{0,1\}^U} (-1)^{\sum_{i \in U} z_i}\, \mathbb{E}[f(z,Y)], \qquad w(U) = b(U)^2, \tag{3.1}$$

where Y has independent Bernoulli-q coordinates on $[N] \setminus U$. In particular $b(\varnothing) = \mu_q(\mathcal{F}) \geq 1/2$.

Step 1: the global bound. We show

$$\sum_{U \subseteq [N]} q^{|U|} w(U) \leq 1. \tag{3.2}$$

(a) Basis. We use the standard biased Fourier basis [7, Section 8.4]:

$$\chi_U(x) = \frac{\prod_{i \in U}(q - x_i)}{[q(1-q)]^{|U|/2}}, \qquad U \subseteq [N].$$

Under μ_q the factors $(q - x_i)/\sqrt{q(1-q)}$ are independent with mean 0 and variance 1, so $\mathbb{E}_{\mu_q}[\chi_U \chi_V]$ equals 1 if $U = V$ and 0 otherwise. Since μ_q has full support, $\mathbb{E}_{\mu_q}[gh]$ is an inner product on the 2^N-dimensional space of real functions on $\{0,1\}^N$, and the 2^N functions χ_U form an orthonormal basis. With $\widehat{f}(U) = \mathbb{E}_{\mu_q}[f\chi_U]$, Parseval gives $\sum_U \widehat{f}(U)^2 = \mathbb{E}_{\mu_q}[f^2]$.

(b) Relation to b. For a Bernoulli-q bit X_i and $z_i \in \{0,1\}$,

$$\mathbb{P}(X_i = z_i)(q - z_i) = q(1-q)(-1)^{z_i},$$

since the left side is $(1-q)q$ for $z_i = 0$ and $q(q-1)$ for $z_i = 1$. Writing the expectation defining $\widehat{f}(U)$ as a sum over $z \in \{0,1\}^U$ and an expectation over the remaining coordinates,

$$\widehat{f}(U) = \frac{[q(1-q)]^{|U|}}{[q(1-q)]^{|U|/2}} \sum_{z \in \{0,1\}^U} (-1)^{\sum_{i \in U} z_i}\, \mathbb{E}[f(z,Y)] = \left(\frac{q}{1-q}\right)^{|U|/2} b(U).$$

(c) Conclusion. Squaring, $q^{|U|} w(U) = (1-q)^{|U|}\, \widehat{f}(U)^2 \leq \widehat{f}(U)^2$. Summing over U and using (a),

$$\sum_U q^{|U|} w(U) \leq \sum_U \widehat{f}(U)^2 = \mathbb{E}_{\mu_q}[f^2] = \mu_q(\mathcal{F}) \leq 1.$$

Step 2: the signed identity. Fix $S \in E_{32}(\mathcal{F})$. We show

$$0 = \sum_{U \subseteq S} (-1)^{|U|} b(U)^{32}. \tag{3.3}$$

(a) A signed measure on arrays. For $y \in \{0,1\}^{32}$ write $|y| = \sum_j y_j$ and

$$P(y) = \prod_{j=1}^{32} q^{y_j}(1-q)^{1-y_j}, \qquad \widetilde{P}(y) = P(y) - (1-q)^{32}(-1)^{|y|}.$$

Then $\widetilde{P}(0,\dots,0) = (1-q)^{32} - (1-q)^{32} = 0$. An array $x \in \{0,1\}^{32\times N}$ has rows $x_1,\dots,x_{32} \in \{0,1\}^N$ and columns $x^1,\dots,x^N \in \{0,1\}^{32}$. Set

$$\nu_S(x) = \prod_{i\in S} \widetilde{P}(x^i) \prod_{i\notin S} P(x^i).$$

(b) Vanishing. If $\nu_S(x) \neq 0$, no column x^i with $i \in S$ is zero, that is, each $i \in S$ belongs to some row. Then $S \subseteq x_1 \cup \cdots \cup x_{32}$, and since $S \in E_{32}(\mathcal{F})$, some row is not in $\mathcal{F}$, so $\prod_j f(x_j) = 0$. Hence every term vanishes in

$$\sum_{x\in\{0,1\}^{32\times N}} \nu_S(x) \prod_{j=1}^{32} f(x_j) = 0. \tag{3.4}$$

Nonnegativity of ν_S is not needed.

(c) Expansion. Multiply out $\prod_{i\in S}[P(x^i) - (1-q)^{32}(-1)^{|x^i|}]$, letting $U \subseteq S$ be the set of columns where the second term is chosen:

$$\nu_S(x) = \sum_{U\subseteq S} (-1)^{|U|}(1-q)^{32|U|} \prod_{i\in U}(-1)^{|x^i|} \prod_{i\notin U} P(x^i).$$

Each factor splits over the rows: $(1-q)^{32|U|} = \prod_j (1-q)^{|U|}$, $\prod_{i\in U}(-1)^{|x^i|} = \prod_j (-1)^{\sum_{i\in U} x_{ji}}$, and $\prod_{i\notin U} P(x^i) = \prod_j \prod_{i\notin U} q^{x_{ji}}(1-q)^{1-x_{ji}}$. So

$$\nu_S(x) = \sum_{U\subseteq S} (-1)^{|U|} \prod_{j=1}^{32} g_U(x_j), \qquad g_U(v) = (1-q)^{|U|}(-1)^{\sum_{i\in U} v_i} \prod_{i\notin U} q^{v_i}(1-q)^{1-v_i}.$$

(d) Row factorization. The rows range independently over $\{0,1\}^N$, so

$$\sum_x \prod_{j=1}^{32} g_U(x_j) f(x_j) = \Big(\sum_{v\in\{0,1\}^N} g_U(v) f(v) \Big)^{32}.$$

Writing $v = (z, y)$, the product over $i \notin U$ in g_U equals $\mathbb{P}(Y = y)$, so the inner sum equals $b(U)$ by (3.1). Substituting (c) and (d) into (3.4) gives (3.3).

Step 3: the local bound. For every $S \in E_{32}(\mathcal{F})$,

$$\sum_{\varnothing\neq U\subseteq S} w(U)^{16} \geq 2^{-32}. \tag{3.5}$$

Indeed, isolate the term $U = \varnothing$ in (3.3), apply the triangle inequality, and use $b(\varnothing) \geq 1/2$:

$$2^{-32} \leq b(\varnothing)^{32} \leq \sum_{\varnothing\neq U\subseteq S} |b(U)|^{32} = \sum_{\varnothing\neq U\subseteq S} w(U)^{16}.$$

Step 4: heavy sets as generators. Set $\rho = q/L$ and

$$\mathcal{G}_0 = \{U \subseteq [N] : U \neq \varnothing,\ w(U) \geq 2^{-64|U|}\}.$$

For $U \in \mathcal{G}_0$ with $r = |U| \geq 1$, since $2^{-64r} \leq w(U)$,

$$\rho^r = q^r 2^{-6r} 2^{-64r} \leq 2^{-6r} q^r w(U) \leq 2^{-6} q^r w(U).$$

Summing and using (3.2),

$$c_\rho(\mathcal{G}_0) \leq 2^{-6}. \tag{3.6}$$

Step 5: bins and the covering lemma. For integers $r \geq 1$ and h, let

$$\mathcal{H}_{r,h} = \{U \subseteq [N] : |U| = r,\ U \notin \mathcal{G}_0, \quad 2^{-(h+1)} < w(U) \leq 2^{-h}\}.$$

Every nonempty $U \notin \mathcal{G}_0$ with $w(U) > 0$ lies in exactly one bin. If $\mathcal{H}_{r,h} \neq \varnothing$, then $h \geq 64r$: a member has $2^{-(h+1)} < w(U) < 2^{-64r}$, so $h + 1 > 64r$. By (3.2),

$$|\mathcal{H}_{r,h}|\, q^r 2^{-(h+1)} < \sum_{U \in \mathcal{H}_{r,h}} q^r w(U) \leq 1,$$

and since $\rho \leq q$,

$$|\mathcal{H}_{r,h}| \rho^r \leq 2^{h+1}.$$

Also $0 < \rho < 1$ and $h \geq 64r \geq \max\{r, 64\}$. So Lemma 2.1, with $X = [N]$ and $\mathcal{H} = \mathcal{H}_{r,h}$, gives a family $\mathcal{G}_{r,h}$ that covers all sets containing at least 2^{12h} members of $\mathcal{H}_{r,h}$ and has

$$c_\rho(\mathcal{G}_{r,h}) \leq 3 \cdot 2^{-h}.$$

Step 6: total cost. Only finitely many bins are occupied, since $2^{[N]}$ is finite. Let

$$\mathcal{G} = \mathcal{G}_0 \cup \bigcup_{\mathcal{H}_{r,h} \neq \varnothing} \mathcal{G}_{r,h}.$$

Since $\sum_{h \geq 64r} 2^{-h} = 2^{1-64r}$ and $\sum_{r \geq 1} 2^{-64r} = 1/(2^{64} - 1)$,

$$3 \sum_{r \geq 1} \sum_{h \geq 64r} 2^{-h} = \frac{6}{2^{64} - 1} < \frac{1}{4}. \tag{3.7}$$

With (3.6) this gives

$$c_\rho(\mathcal{G}) \leq \frac{1}{64} + \frac{6}{2^{64} - 1} < \frac{1}{2}.$$

Step 7: $\mathcal{G}$ covers $E_{32}(\mathcal{F})$. Suppose $S \subseteq [N]$ contains no member of $\mathcal{G}$. Then no nonempty $U \subseteq S$ is in $\mathcal{G}_0$, so each such U has $w(U) = 0$ or lies in a bin. Also S contains fewer than 2^{12h} members of each occupied bin $\mathcal{H}_{r,h}$, each with $w(U)^{16} \leq 2^{-16h}$. Hence

$$\sum_{\varnothing \neq U \subseteq S} w(U)^{16} \leq \sum_{r \geq 1} \sum_{h \geq 64r} 2^{12h} 2^{-16h} = \sum_{r \geq 1} \frac{2^{-256r}}{1 - 2^{-4}} = \frac{16}{15(2^{256} - 1)} < 2^{-32}. \tag{3.8}$$

By (3.5), $S \notin E_{32}(\mathcal{F})$. So every member of $E_{32}(\mathcal{F})$ contains a member of $\mathcal{G}$, and $c_{q/L}(\mathcal{G}) < 1/2$ by Step 6. □

4 Returning to the original density

Plan. We remove the factor L from Proposition 3.1 by coupling L dependent random sets. Each has exactly the law μ_p, and their union has a larger, exactly known density. Since each row is controlled only through its own law, a union bound suffices, and arbitrary $\mathcal{D}$ are allowed.

Proof of Theorem 1.1. Set $L = 2^{70}$ and $k = 32L = 2^{75}$, and suppose $\mu_p(\mathcal{D}) \geq 1 - 1/k$.

Step 0: union bound. Let $R_1, \ldots, R_L$ be random subsets of $[N]$ on a common probability space, each with law μ_p; no independence between them is assumed. Then

$$\mathbb{P}(R_j \in \mathcal{D} \text{ for all } 1 \leq j \leq L) \geq 1 - \sum_{j=1}^{L} \mathbb{P}(R_j \notin \mathcal{D}) \geq 1 - \frac{L}{k} = \frac{31}{32}. \tag{4.1}$$

Case 1: $Lp < 1$**.**

(a) Coupling. Independently for each $i \in [N]$, choose a label $J_i \in \{0, 1, \ldots, L\}$ with

$$\mathbb{P}(J_i = 0) = 1 - Lp, \qquad \mathbb{P}(J_i = j) = p \quad (1 \leq j \leq L),$$

and put $R_j = \{i : J_i = j\}$. For fixed j, the events $\{i \in R_j\}$ are independent over i, each of probability p, so R_j has law μ_p. The union $R = R_1 \cup \cdots \cup R_L$ equals $\{i : J_i \neq 0\}$, so it has law μ_q with $q = Lp \in (0, 1)$.

(b) The family $\mathcal{F}$*.* Define

$$\mathcal{F} = \{D_1 \cup \cdots \cup D_L : D_1, \ldots, D_L \in \mathcal{D}\}.$$

On the event in (4.1), $R \in \mathcal{F}$. Hence

$$\mu_q(\mathcal{F}) = \mathbb{P}(R \in \mathcal{F}) \geq \frac{31}{32} \geq \frac{1}{2}.$$

(c) Exceptional families agree. A union of 32 members of $\mathcal{F}$ is a union of $32L$ members of $\mathcal{D}$. Conversely, any $32L$ members of $\mathcal{D}$, grouped into 32 blocks of L, give 32 members of $\mathcal{F}$ with the same union. So the unions allowed in (1.1) are the same, and

$$E_{32}(\mathcal{F}) = E_{32L}(\mathcal{D}) = E_k(\mathcal{D}).$$

(d) Conclusion. By Proposition 3.1, $E_{32}(\mathcal{F})$ is (q/L)-small, and $q/L = p$.

Case 2: $Lp \geq 1$**.** Set

$$\tau = \frac{p - 1/L}{1 - 1/L} \in [0, 1).$$

Independently for each coordinate i: choose a *mandatory row* uniformly among the L rows and include i in it; conditionally on this choice, include i in each other row independently with probability τ. For each fixed j,

$$\mathbb{P}(i \in R_j) = \frac{1}{L} + \left(1 - \frac{1}{L}\right)\tau = p,$$

and these events are independent over i. So each R_j has law μ_p, while $R_1 \cup \cdots \cup R_L = [N]$ in every outcome.

By (4.1), some outcome has all L rows in $\mathcal{D}$. Repeating this tuple 32 times gives k members of $\mathcal{D}$ with union $[N]$. So no $S \subseteq [N]$ is exceptional: $E_k(\mathcal{D}) = \varnothing$, which the empty family covers at cost 0. The boundary $Lp = 1$ is included, with $\tau = 0$. □

5 Verification checklist

Every step of Sections 2–4 was rechecked while rewriting, and none is marked unverified. The only statements taken on trust are the three external results in the proof of Corollary 1.2, which is not used elsewhere. A sceptical reader should check these first:

1. **Extension bound** (2.3) (Lemma 2.1, Step 2): each class $\{e \cap B = I\}$ contributes at most $d(I) \le T$, and this uses regularity through (2.2).
2. **Residual cost** (Lemma 2.1, Step 5): the recursion $W_{t+1} \le T(1+rm)^r W_t$, the independence of the m^2 pair indices, and the inequality $4h + 13rh + 1 \le \pi m$.
3. **Normalization** $\widehat{f}(U) = (q/(1-q))^{|U|/2} b(U)$ (Proposition 3.1, Step 1(b)), which turns Parseval into (3.2).
4. **Signed identity** (3.3) (Step 2): $\widetilde{P}(0) = 0$ forces the rows to cover S, and the expansion factors over rows into $b(U)^{32}$.
5. **Bin bookkeeping** (Steps 5–7): occupied bins have $h \ge 64r$, the hypotheses of Lemma 2.1 hold with $\rho = q/2^{70}$, and the sums (3.7) and (3.8) fall below $1/4$ and 2^{-32}.
6. **Couplings** (Section 4): each row has law μ_p, the union has law μ_{Lp} (Case 1) or equals $[N]$ (Case 2), and $E_{32}(\mathcal{F}) = E_k(\mathcal{D})$.

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